Supplementary materials: Context-Aware Subgraph Explanations for Multi-Task GNNs

A CAMIE Algorithm Details

Algorithm 1 summarizes the overall procedure for both training the shared context-aware motif scorer and ranking motifs for a query molecule–assay pair.

Algorithm 1 Training and Motif Ranking of CAMIE

Require: Training dataset $\mathcal{D}$; query pair (G, a_j); motif sets $\{\mathcal{M}_{G^{(i)}}\}_{i=1}^{N}$ and $\mathcal{M}_G$; motif partitions $\{\mathcal{P}_{G^{(i)}}\}_{i=1}^{N}$ and $\mathcal{P}_G$; frozen encoder f_ϕ; readout function r; frozen output head f_o; assay-context encoder $g(\cdot;\eta)$; context-conditioned motif scorer q_θ; number of selected motifs k

Ensure: Top-k context-aware motif explanation $\mathcal{E}_k(G, a_j)$

for $i = 1$ to N **do**
    $H_{G^{(i)}} \leftarrow f_\phi(X^{(i)}, A^{(i)})$
    $z_{G^{(i)}} \leftarrow r(H_{G^{(i)}})$
    for all $m_s \in \mathcal{M}_{G^{(i)}}$ with node index set $I_s \in \mathcal{P}_{G^{(i)}}$ **do**
        $z_{m_s}^{(i)} \leftarrow \frac{1}{|I_s|} \sum_{v \in I_s} h_v^{(i)}$
        Define $X_{\setminus m_s}^{(i)}$ by $(X_{\setminus m_s}^{(i)})_v \leftarrow \mathbf{1}[v \notin I_s] X_v^{(i)}$
        $H_{G^{(i)} \setminus m_s} \leftarrow f_\phi(X_{\setminus m_s}^{(i)}, A^{(i)})$
        $z_{G^{(i)} \setminus m_s} \leftarrow r(H_{G^{(i)} \setminus m_s})$
        for all $a_j \in \mathcal{T}^{(i)}$ **do**
            $z_{a_j} \leftarrow g(a_j; \eta)$
            $z'_{a_j} \leftarrow W_a z_{a_j} + b_a$
            $u_{s,j}^{(i)} \leftarrow [z_{m_s}^{(i)}; z_{G^{(i)}}; z'_{a_j}; z_{m_s}^{(i)} \odot z_{G^{(i)}}; z_{m_s}^{(i)} \odot z'_{a_j}]$
            $\hat{t}_{s,j}^{(i)} \leftarrow q_\theta(u_{s,j}^{(i)})$
            $t_{s,j}^{(i)} \leftarrow \left| F(G^{(i)})_j - F(G^{(i)} \setminus m_s)_j \right|$
        end for
    end for
end for
$\theta^* \leftarrow \arg\min_\theta \mathcal{L}(\mathcal{D}; \theta)$
Compute $H_G \leftarrow f_\phi(X, A)$, $z_G \leftarrow r(H_G)$, and $z_{a_j} \leftarrow g(a_j; \eta)$
for all $m_s \in \mathcal{M}_G$ with node index set $I_s \in \mathcal{P}_G$ **do**
    $z_{m_s} \leftarrow \frac{1}{|I_s|} \sum_{v \in I_s} h_v$
    $z'_{a_j} \leftarrow W_a z_{a_j} + b_a$
    $u_{s,j} \leftarrow [z_{m_s}; z_G; z'_{a_j}; z_{m_s} \odot z_G; z_{m_s} \odot z'_{a_j}]$
    $\hat{t}_{s,j} \leftarrow q_{\theta^*}(u_{s,j})$
end for
$\mathcal{E}_k(G, a_j) \leftarrow \mathrm{Top}_k \left(\{(m_s, \hat{t}_{s,j})\}_{s=1}^{M} \right)$
return $\mathcal{E}_k(G, a_j)$

B Detailed Sensitivity Analysis

Table 1 provides the full numeric results for the top-k sensitivity analysis across all evaluated post-hoc explainers.

Table 1: Top-k sensitivity analysis. Values are reported as mean $\pm$ std across seeds.

Model	k	$\texttt{Fidelity}^{+}_{F1}$ ($\uparrow$)	$\texttt{Fidelity}^{-}_{F1}$ ($\downarrow$)
CAMIE (ours)	1	$\mathbf{0.2774 \pm 0.0899}$	0.1593 ± 0.0610
	2	$\mathbf{0.3024 \pm 0.0924}$	0.0945 ± 0.0458
	3	$\mathbf{0.3076 \pm 0.0900}$	0.0554 ± 0.0373
	4	$\mathbf{0.3078 \pm 0.0884}$	0.0369 ± 0.0337
	5	$\mathbf{0.3077 \pm 0.0877}$	0.0228 ± 0.0244
eXEL	1	0.2635 ± 0.0883	0.1617 ± 0.0609
	2	0.2913 ± 0.0918	0.1053 ± 0.0466
	3	0.2956 ± 0.0901	0.0751 ± 0.0375
	4	0.2985 ± 0.0879	0.0601 ± 0.0324
	5	0.2987 ± 0.0880	0.0470 ± 0.0272
SA	1	0.1358 ± 0.0458	0.1756 ± 0.0777
	2	0.1861 ± 0.0712	0.1540 ± 0.0728
	3	0.2178 ± 0.0880	0.1235 ± 0.0630
	4	0.2432 ± 0.1012	0.0981 ± 0.0541
	5	0.2582 ± 0.1048	0.0698 ± 0.0459
GBP	1	0.1685 ± 0.0307	0.1913 ± 0.0572
	2	0.2405 ± 0.0546	0.1622 ± 0.0505
	3	0.2746 ± 0.0685	0.1206 ± 0.0415
	4	0.2924 ± 0.0779	0.0888 ± 0.0365
	5	0.3018 ± 0.0828	0.0611 ± 0.0343
GradCAM	1	0.1910 ± 0.0323	0.1781 ± 0.0616
	2	0.2455 ± 0.0534	0.1437 ± 0.0530
	3	0.2723 ± 0.0664	0.1052 ± 0.0438
	4	0.2864 ± 0.0753	0.0739 ± 0.0355
	5	0.2942 ± 0.0784	0.0512 ± 0.0325
PGExplainer	1	0.1654 ± 0.0641	0.1930 ± 0.0639
	2	0.2323 ± 0.0913	0.1407 ± 0.0551
	3	0.2623 ± 0.0929	0.1041 ± 0.0444
	4	0.2767 ± 0.0919	0.0765 ± 0.0334
	5	0.2874 ± 0.0907	0.0559 ± 0.0262
GNNExplainer	1	0.1045 ± 0.0194	0.2091 ± 0.0637
	2	0.1566 ± 0.0385	0.1912 ± 0.0637
	3	0.2006 ± 0.0515	0.1639 ± 0.0591
	4	0.2362 ± 0.0627	0.1337 ± 0.0494
	5	0.2596 ± 0.0705	0.1046 ± 0.0416

C Additional Experimental Analysis

C.1 Assay-wise explanation diversity

To quantify whether the selected motif set changes across assay contexts for the same molecule, we compare the top-k selected motif sets S^+ across all assay pairs associated with each molecule. For each assay pair, we compute the Jaccard similarity between the two selected sets. We further summarize the fraction of assay pairs that are exact matches, partially overlapping, or non-overlapping.

Table 2 shows that CAMIE exhibits lower average Jaccard similarity and fewer exact matches than Graph-only, indicating that its selected motif sets vary more across assays for the same molecule. Compared with Context-only, CAMIE also produces a larger fraction of partially overlapping and non-overlapping cases, suggesting that assay context is used to adapt motif selection rather than producing a fixed molecule-specific explanation. By contrast, Graph-only always returns the same selected motif set across assays, confirming that without assay context the scorer collapses to assay-invariant explanations.

Table 2: Frequency of S^+ changes across assay pairs for the same molecule.

Method	Mean Jaccard	Exact Match	Partial Overlap	No Overlap
CAMIE (ours)	0.717	0.613	0.313	0.074
Context-only	0.767	0.678	0.262	0.059
Graph-only	1.000	1.000	0.000	0.000

C.2 Assay-wise comparison with ST-MIE

We further compare CAMIE with ST-MIE at the assay level. Table 3 summarizes win/tie/loss statistics and fidelity gaps across the 30 assays. CAMIE outperforms ST-MIE on the majority of assays and shows a positive mean gap, indicating that the shared context-conditioned scorer provides more consistent assay-level gains.

Table 3: Assay-level comparison of $\texttt{Fidelity}^+_{F1}$. Gap is CAMIE minus ST-MIE. Ties include assays with $|\text{Gap}| < 0.005$ or non-positive $\texttt{Fidelity}$ for both methods.

Method	Win / Tie / Loss	Mean Gap	Median Gap	Max Gap
CAMIE vs. ST-MIE	21/6/3	+0.021	+0.021	+0.090

Figure 1 further stratifies this comparison by assay resource level. The advantage of CAMIE is more pronounced in low-resource assay groups, which supports the interpretation that cross-assay sharing is particularly beneficial when assay-specific data are limited. Table 4 provides the full numeric details for each assay sorted by sample size.

Table 4: Per-assay $\texttt{Fidelity}^{+}_{F1}$ comparison between ST-MIE and CAMIE on selected assays, sorted by sample size.

Group	Assay	#Samples	CAMIE	ST-MIE
Low	NVS_NR_rMR	77	0.7586 ± 0.2686	0.6689 ± 0.3429
	NVS_ADME_hCYP1A1	137	0.3095 ± 0.2319	0.2746 ± 0.2077
	NVS_NR_hTRa_Antagonist	187	0.2632 ± 0.0827	0.2430 ± 0.0499
	NVS_NR_hAR	189	0.4599 ± 0.2464	0.4820 ± 0.2291
	NVS_ADME_hCYP19A1	198	0.4447 ± 0.1395	0.4035 ± 0.1734
	NVS_NR_rAR	223	0.3678 ± 0.2414	0.3680 ± 0.2253
	NVS_NR_cAR	248	0.5676 ± 0.3267	0.5039 ± 0.3463
	NVS_NR_hGR	282	0.2642 ± 0.2367	0.2375 ± 0.2237
	CEETOX_H295R_OHPREG_up	364	0.4270 ± 0.0822	0.4472 ± 0.0825
	CEETOX_H295R_PROG_up	371	0.4227 ± 0.1156	0.4088 ± 0.1084
	Avg.		0.4285 (+0.0248)	0.4037
Medium	CEETOX_H295R_OHPROG_up	390	0.4290 ± 0.1168	0.4250 ± 0.1028
	CEETOX_H295R_OHPROG_dn	390	0.3382 ± 0.2147	0.2989 ± 0.2323
	CEETOX_H295R_ESTRONE_up	390	0.1976 ± 0.1581	0.2166 ± 0.1490
	CEETOX_H295R_ESTRONE_dn	390	0.1859 ± 0.1251	0.1626 ± 0.1368
	CEETOX_H295R_CORTISOL_dn	390	0.4547 ± 0.1830	0.4054 ± 0.2061
	CEETOX_H295R_11DCORT_dn	390	0.3268 ± 0.2400	0.2833 ± 0.2411
	NVS_NR_hER	945	0.3752 ± 0.1350	0.3591 ± 0.1327
	OT_AR_ARSRC1_0960	1461	0.1793 ± 0.1166	0.1445 ± 0.1490
	OT_ER_ERbERb_0480	1467	0.4484 ± 0.0569	0.4269 ± 0.0807
	OT_ER_ERaERb_0480	1477	0.4808 ± 0.0503	0.4714 ± 0.0525
	Avg.		0.3416 (+0.0222)	0.3194
High	ATG_GR_TRANS_up	2938	0.1367 ± 0.1034	0.1282 ± 0.0995
	ATG_GR_TRANS_dn	2938	0.0000 ± 0.0000	0.0000 ± 0.0000
	ATG_ERa_TRANS_up	2938	0.3985 ± 0.0684	0.3762 ± 0.0845
	ATG_AR_TRANS_dn	2938	-0.0119 ± 0.0315	-0.0104 ± 0.0295
	ATG_AR_TRANS_up	2938	0.3027 ± 0.0857	0.2945 ± 0.0840
	ATG_ERE_CIS_dn	2938	-0.0156 ± 0.0413	-0.0125 ± 0.0354
	ATG_ERE_CIS_up	2938	0.3507 ± 0.0805	0.3258 ± 0.1056
	TOX21_VDR_BLA_agonist_ratio	5872	-0.0123 ± 0.0273	-0.0124 ± 0.0254
	TOX21_TR_LUC_GH3_Antagonist	6353	0.1978 ± 0.1724	0.1467 ± 0.2152
	TOX21_AR_LUC_MDAKB2_Agonist	6353	0.4470 ± 0.0585	0.4018 ± 0.1369
	Avg.		0.1793 (+0.0156)	0.1638

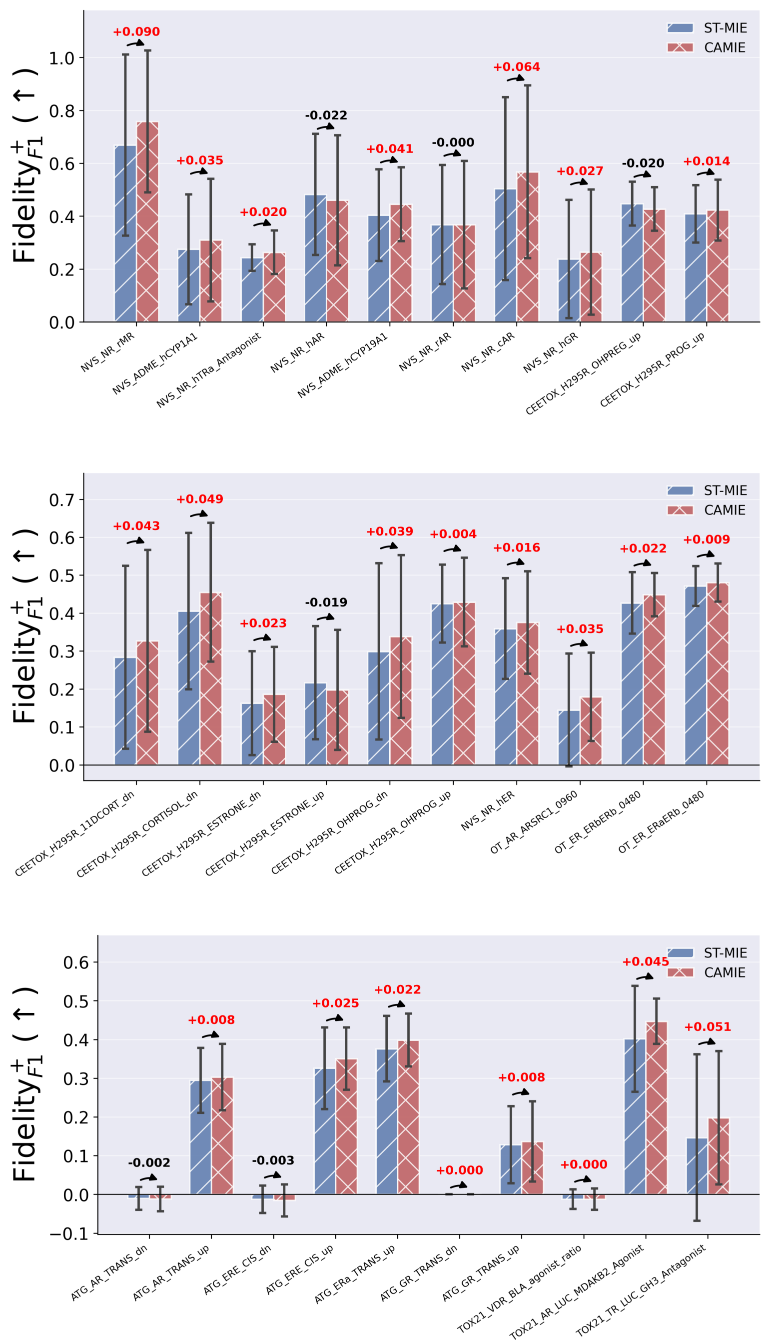

Figure 1: Model analysis for assay context pair resources. Assays are grouped by the number of training molecule-assay pairs (Low, Medium, High). The means and their standard errors across 10 random seeds are reported. ↑ denotes that higher is better.

C.3 Relationship to assay-level structure

To examine whether CAMIE explanations reflect assay-level structure beyond predictive faithfulness, we analyze two complementary aspects.

First, we compare the overlap of selected motif sets across metadata-defined assay groupings. Table 5 reports the mean S^+ Jaccard overlap for assay pairs belonging to the same versus different groups defined by assay domain, tier, mechanism of action (MoA), and target gene. Across all groupings, assay pairs within the same metadata group show higher overlap, with the largest gap observed for MoA. This suggests that CAMIE explanations partially align with biologically meaningful assay organization.

Table 5: Mean S^+ Jaccard overlap for same vs. different metadata-defined assay groups.

Grouping	Same Group	Different Group	Gap
Domain	0.606	0.583	+0.023
Tier	0.629	0.583	+0.046
MoA	0.658	0.586	+0.073
Target gene	0.639	0.587	+0.052

Second, we stratify assay pairs by label agreement. Table 6 shows that assay pairs with different labels exhibit slightly larger score distances and lower S^+ overlap than label-matched assay pairs. This indicates that explanation differences are not arbitrary, but tend to increase when the assay-specific predictive outcomes diverge.

Table 6: Explanation shift by label agreement. Label-different assay pairs tend to show slightly larger score distance and lower S^+ overlap.

Subset	Mean Score Dist.	Mean S^+ Jaccard	Exact Match	No Overlap
All	0.079	0.595	0.452	0.119
Label same	0.077	0.600	0.458	0.115
Label different	0.086	0.577	0.430	0.129